Beyond Task Completion: A Cumulative Arithmetic State-Tracking Probe Provides Incremental Prediction on a Deterministic Agent Battery Across Open-Weight Models

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Abstract

Task-completion metrics dominate LLM agent evaluation, but they may not capture the process-level competencies that underlie reliable multi-step behavior—a gap we call the Completion Fallacy. We introduce WMF-AM, a probe of cumulative arithmetic state tracking under load, and evaluate it on 20 open-weight models from 13 families (0.5B–35B parameters). In a pre-specified confirmatory analysis, WMF-AM predicts performance on an independent released deterministic 10-task agent battery: Kendall's τ=0.612 (p<0.001, bootstrap 95% CI $[0.360, 0.814]$). Confirmatory template stability holds across prompt formats (τ=0.524, p=0.006). Targeted controls and ablations establish the narrower construct: WMF-AM difficulty is not explained by single-step arithmetic, generic entity tracking, or per-operation arithmetic parsing. Exploratory (hypothesis-generating) analyses suggest the predictive signal may extend beyond completion score and model scale, and also appears in a subset of tasks annotated by the authors as not centrally involving state tracking (τ=0.623); all exploratory results require held-out replication. These results demonstrate that a process-sensitive probe captures evaluation-relevant variance a completion battery misses, within this sample of open-weight models. Generalization to closed API models remains open.

1 Introduction

This paper presents a predictive-validity study for a process-sensitive probe. It does not propose replacing task-completion evaluation; it asks whether a controlled probe of cumulative arithmetic state tracking provides incremental predictive value for downstream agent performance beyond completion. Task-completion metrics remain useful and widely applicable. Their limitation, documented here and in prior work, is that completion scores may compress variance in process-level capabilities that matter for reliable multi-step agent behavior—a pattern we call the Completion Fallacy: the inference that completion success alone is sufficient for inferring latent process competence. Benchmarks such as HotpotQA [Yang et al., 2018], MMLU [Hendrycks et al., 2021a], and AgentBench [Liu et al., 2023] measure what LLMs accomplish; process-sensitive probes ask how they process information under load, as a complementary—not competing—evaluation lens.

Several recent results converge on this point. PAE [Xie et al., 2026] finds 27–78% of agent successes involve corrupted trajectories; Fragile Thoughts [Aravindan and Kejriwal, 2026] and Robust Answers Fragile Logic [Jiang et al., 2026] show correct answers produced despite broken reasoning chains; Turpin et al. [Turpin et al., 2023] demonstrate unfaithful chain-of-thought explanations. These are instances of multiple realizability [Putnam, 1967]: identical outcomes produced by categorically different processes—a pattern also documented as underspecification [D'Amour et al., 2022] and shortcut learning [Geirhos et al., 2020].

This paper introduces and empirically validates WMF-AM, a controlled probe for cumulative arithmetic state tracking under load. The WMF-AM validation is the sole empirical contribution. We situate it within an interpretive taxonomy (CEF) that provides nomenclature and motivation for this and related probes; CEF is explicitly a working conceptual taxonomy, not a validated empirical contribution of this paper. The probe is conceptually related to entity tracking [Kim and Schuster, 2023] and latent state persistence [Anonymous, 2025]; it is distinguished by cumulative arithmetic load manipulation and downstream predictive validity on a deterministic agent battery.

This paper makes four contributions. (1) Measurement construct and probe. We operationalize cumulative arithmetic state tracking under load as a measurable process capability and introduce WMF-AM, a controlled probe for testing it (Section 3). (2) Confirmatory predictive validity. Across 20 open-weight models from 13 families, WMF-AM predicts performance on an independent released deterministic battery (τ=0.612, p<0.001, 95% CI $[0.360, 0.814]$; pre-specified, Bonferroni-corrected confirmatory, Section 4). (3) Confirmatory template stability (τ=0.524, p=0.006; pre-specified). Exploratory (hypothesis-generating only): partial τ analyses suggest the predictive relationship survives controls for task completion (partial τ=0.411) and model scale (partial τ=0.503); these require held-out replication and should not be treated as confirmatory. Non-arithmetic ablation, K=1 control, and yoked cancellation confirm the signal is not explained by single-step arithmetic or generic tracking. (4) Robustness. The signal is stable across seeds, templates, paraphrases, and leave-one-out analyses (Section 4). A glossary of all acronyms appears in Table 10 (Appendix C).

Scope. The empirical contribution is the introduction and validation of WMF-AM as one process-sensitive probe of cumulative arithmetic state tracking, evaluated on 20 open-weight models. The two pre-specified confirmatory analyses are predictive validity (τ=0.612) and template stability (τ=0.524); all other results are exploratory. Generalization to closed API models, additional downstream criteria, and broader model families requires held-out evaluation. The broader CEF taxonomy is an interpretive framework, not an empirical result of this paper.

Paper roadmap. Section 2 formalizes the problem and reviews related work; Section 3 introduces WMF-AM and the CEF framework; Section 4 reports the validation study; Section 5 discusses limitations and implications. Table 8 in Appendix B gives development status for all 11 CEF sub-dimensions.

2 The Completion Fallacy and Related Work

Three failure modes of completion-only evaluation. (F1) Diagnostic blindness: models with near-identical completion scores can have very different process profiles. In the study (Section 4): 30/105 model pairs with $|\Delta\text{completion}| \leq 0.05$ show $|\Delta\text{WMF-AM}| \geq 0.10$ (exploratory; at $\leq 0.03/ \geq 0.15$, 18/105 pairs qualify). WMF-AM spans 0.050–0.983 among 20 models while completion spans only 0.44–0.92; among mid-to-large models ($\geq$3B), completion SD = 0.052 versus WMF-AM SD = 0.228. (F2) Uninformative failure diagnosis: current benchmarks report that an agent failed, not why—whether the bottleneck is state tracking, metacognitive calibration, or episodic confusion. Process-sensitive profiles could in principle decompose failure into actionable sub-dimensions; this requires validation at scale. (F3) Safety blindness: pattern-matching agents (which fail predictably at distribution shift)

are indistinguishable from genuine reasoners on standard benchmarks. Whether process probes predict this failure mode remains an open empirical question.

Empirical support. Multiple realizability is empirically the norm in frontier LLMs. Fragile Thoughts [Aravindan and Kejriwal, 2026] and Robust Answers Fragile Logic [Jiang et al., 2026] provide direct evidence: correct answers are regularly produced despite corrupted reasoning steps. Turpin et al. [Turpin et al., 2023] show chain-of-thought explanations can be systematically unfaithful; Lanham et al. [Lanham et al., 2023] measure this faithfulness gap directly. Uesato et al. [Uesato et al., 2022] demonstrate that process-based feedback outperforms outcome-based feedback for mathematical reasoning. PAE [Xie et al., 2026] extends this to agent trajectories, finding 27–78% of successful completions involve corrupted intermediate actions. These are Agent B/C behaviors at scale.

From outcome underdetermination to process probes. Outcome metrics cannot distinguish process quality. Three questions follow: (a) What can task success not identify? Whether behavior generalizes under shift, whether confidence reflects error detection, or whether memory discriminates episodes. (b) Which gaps require process evidence versus robustness or adversarial testing? Process probes are needed when the question is how the model processes information. (c) Why CEF? Robustness testing [Ribeiro et al., 2020] addresses perturbation; trace audits [Turpin et al., 2023, Lanham et al., 2023] address reasoning-to-output coupling; adversarial evaluations address worst-case performance. CEF targets the space left uncovered: parametrically controlled, cognitively grounded probes of process quality under multi-turn load.

Positioning. Cognitive science has developed rigorous measurement paradigms for each function CEF addresses: working memory capacity [Miller, 1956, Baddeley and Hitch, 1974], metacognitive monitoring/control [Flavell, 1979, Nelson and Narens, 1990], episodic memory [Tulving et al., 1972, Johnson et al., 1993], and cognitive load adaptation [Sweller, 1988, Miyake et al., 2000]. CEF adapts their measurement logic for API-level LLM evaluation. This does not require LLMs to have human cognition; it requires that the constructs pick out capabilities that matter for deployment. NeuroCognition [Haznitrama et al., 2026] measures cognitive ability; CEF measures process quality under load. Strong Memory, Weak Control [de Langis et al., 2025] studies executive function in isolated tasks; CEF tests the same constructs under sustained interference. Entity Tracking [Kim and Schuster, 2023] is prior art for WMF-AM's design; Latent State Persistence [Anonymous, 2025] probes sequential latent-state manipulation via structural operations (rotation, removal) on hidden integers—a closely related paradigm. WMF-AM differs in three ways: (a) operations are cumulative arithmetic (addition/subtraction on named entities), not structural rearrangements; (b) a targeted ablation suite (yoked cancellation, K=1 control, non-arithmetic ceiling) establishes the narrower construct; (c) the study adds downstream predictive validity on a deterministic agent battery. Intelligence Degradation [Wang et al., 2026] is prior art for CLA. Bowman and Dahl [Bowman and Dahl, 2021] argue benchmarking requires construct-level validity; CheckList [Ribeiro et al., 2020] introduced behavioral testing templates; CEF extends both. Process supervision [Lightman et al., 2023, Uesato et al., 2022] trains models using step-level labels; CEF measures process quality but does not prescribe training signals. Mechanistic interpretability analyzes internal representations; CEF operates at the API level (prompt/response only) and is complementary: interpretability explains why a process score is low, while CEF identifies that it is low. A full related-work table appears in Appendix A.

Formal definitions. Let $Y_i \in [0, 1]$ denote the outcome score and $Z_i \in [0, 1]$ a process-sensitive measure for model i, both defined relative to a specific benchmark suite and model population. We say outcome metrics are statistically insufficient with respect to a downstream criterion D if process measures provide significant incremental prediction of D beyond completion: partial $\tau(Z, D \mid Y) > 0$ at conventional α. This is a benchmark-conditional claim—it asserts insufficiency for a given $\langle Y, Z, D\rangle$ triple, not a universal property. Statistical insufficiency does not by itself determine practical or deployment rele-

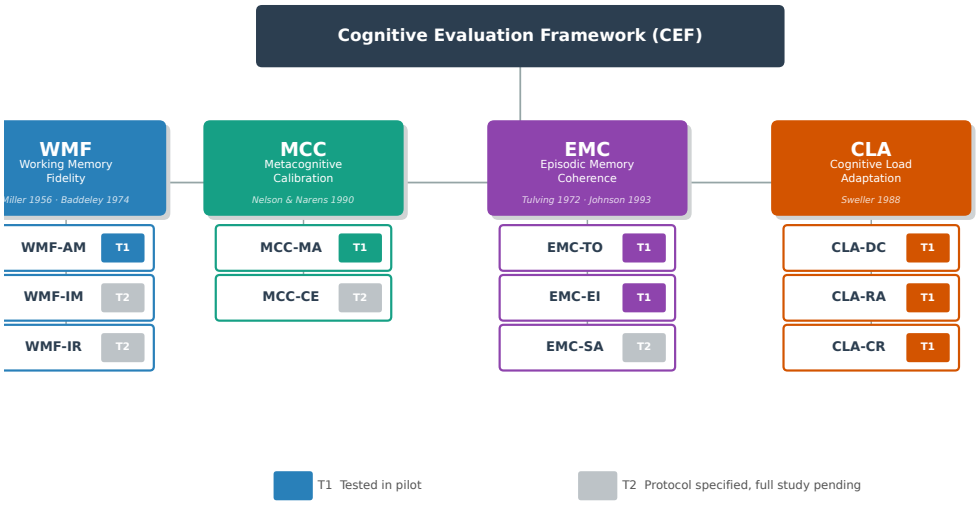

Figure 1. CEF architecture. Four dimensions (WMF, MCC, EMC, CLA) derived from cognitive science, decomposed into 11 proposed sub-dimensions. Solid border: pilot-supported (WMF-AM). Dashed border: preliminary or proposed probes awaiting the $N \geq 15$ study.

vance: the latter depends on whether the incremental signal is large enough and the criterion important enough to warrant the additional measurement cost. In the study, partial $\tau(\text{WMF-AM}, \text{Agent} \mid Y) = 0.411$ ($p = 0.011$, N=20), meeting this criterion. The completion–process correlation ($\tau_{\text{WMF-AM, completion}}$=0.279, p=0.150, N=15, bootstrap 95% CI $[-0.156, 0.637]$) is suggestive of divergence but not confirmed at this sample size. Falsification: CEF's core claim would be undermined if, across multiple probe families and downstream criteria, process measures never provided significant incremental prediction beyond completion. The four-dimension structure is a working taxonomy; dimensional separability requires factor analysis at $N \geq 30$.

3 WMF-AM and Its Interpretive Framework (CEF)

Important framing: This section serves two distinct purposes. (A) It defines WMF-AM, the sole empirical contribution of this paper. (B) It situates WMF-AM within an interpretive taxonomy (CEF) that provides organizing vocabulary; CEF is not a validated empirical contribution. Readers primarily interested in the empirical result can focus on the WMF-AM description (Flagship probe subsection) and skip the broader CEF architecture.

CEF is a working conceptual taxonomy—a design lens, not a validated psychometric structure—grounded in cognitive science. We present it solely as background context for why WMF-AM was designed the way it was, and as a research agenda for future probes. The dimensional structure (4 dimensions, 11 sub-dimensions) is hypothesized, pending factor analysis at $N \geq 30$, and is not presented as an empirical result of this paper. Construct scope: the current evidence supports the narrower construct of cumulative arithmetic state tracking under load, not broad working-memory capacity; the WMF label is operational shorthand (see Appendix D). WMF-AM has comprehensive validation support (predictive validity, seed reliability, template stability, yoked control, arithmetic-vs-assignment ablation, paraphrase stability, K=1 arithmetic control); three other sub-dimensions have preliminary single-seed results; the remainder are protocol-only (Table 8). The number and boundaries of CEF dimensions are working hypotheses pending factor analysis at $N \geq 30$. A full glossary of acronyms (WMF, MCC, EMC, CLA, and their sub-dimension suffixes) appears in Table 10 (Appendix C).

Design principles. Each dimension must have (i) a cognitive science anchor—an established measurement paradigm with validated human operationalizations, (ii) a documented evalu-

Table 1. CEF dimensions: target, failure signature, and exclusions. Detailed sub-dimension definitions appear in Appendix E.

Dim	Target	Failure signature	Excluded
WMF	Active maintenance and transformation of state under load	Accuracy drops as concurrent items exceed threshold	Long-context retrieval; knowledge recall
MCC	Self-monitoring of errors (MA) and correction of flagged errors (CE)	Confident-wrong; flagging without correction	General calibration (ECE); cross-model detection
EMC	Temporal ordering, cross-episode discrimination, source attribution	Episode blending; source misattribution	Single-turn retrieval; semantic memory
CLA	Degradation shape and resource scaling under load	Cliff-edge failure; non-recovery	Knowledge difficulty; unrelated complexity

ation gap not covered by existing benchmarks, and (iii) API-level testability requiring only prompt/response access. We use labels like "working memory" as operational shorthand for behavioral probes, not as claims about LLM mechanisms. "WMF-AM" means accuracy on sequential state-tracking tasks degrades with depth K; it does not mean LLMs have a working memory buffer (see Appendix D for full discussion).

Four dimensions. Table 1 summarizes each dimension's target capacity, canonical failure signature, and exclusions.

Flagship probe: WMF-AM. WMF-AM (Active Manipulation) is a probe of cumulative arithmetic state tracking under load, inspired by working-memory span paradigms. An initial numeric state is presented, K sequential additive/subtractive operations arrive, and the model must report the final cumulative state. The pilot uses depths $K \in \{3, 5, 7\}$ (dissociation study) and $K \in \{2, 4, 6, 8, 12\}$ (yoked cancellation control), across three surface forms (points scoring, warehouse inventory, bank accounts). Prior entity tracking [Kim and Schuster, 2023] tests passive retrieval; WMF-AM tests sequential arithmetic transformation under load. Scope of the construct: WMF-AM specifically measures difficulty with maintaining a running cumulative sum across K additive/subtractive operations. It does not claim to measure general working-memory capacity, and it may partially reflect standalone arithmetic competence; a completed last-operation-only (K=1) control addresses this: 13/15 models achieve near-ceiling single-step arithmetic accuracy ($\bar{x}$=0.957) versus $\bar{x}$=0.151 at K=7, indicating that cumulative accumulation under load—not arithmetic skill per se—is the primary difficulty driver.

Worked example (K=3, points form): "Alice starts with 10 points. Alice gains 5 points. Alice loses 3 points. Alice gains 7 points. What is Alice's current score?" Correct answer: 19. Scoring: exact match (1 if the model outputs 19, 0 otherwise). A model that tracks each update cumulatively ($10 \to 15 \to 12 \to 19$) succeeds; a model that performs only the last operation or guesses a plausible number fails. At K=7, the chain length taxes state maintenance, producing the diagnostic depth×accuracy interaction. Yoked control example (K=3): "Alice starts with 10 points. Alice gains 5 points. Alice loses 5 points. Alice gains 3 points. Alice loses 3 points. Alice gains 7 points. Alice loses 7 points. What is Alice's current score?" Correct answer: 10 (initial state). Each operation is immediately followed by its inverse, requiring identical arithmetic but no cumulative tracking.

Table 2. Metacognitive profile taxonomy. Each cell implies a different intervention target.

	High MCC-CE	Low MCC-CE
High MCC-MA	Genuine metacognition	Monitoring works; correction broken
Low MCC-MA	Correction works; detection unreliable	No metacognitive capability

MCC: Metacognitive Calibration. MCC decomposes metacognition into monitoring accuracy (MCC-MA: predict which of 20 answers are wrong; Pearson r) and control efficacy (MCC-CE: flag uncertain answers, then revise them; decomposed into flagging rate, correction efficacy, and false alarm rate)—Reflexion [Shinn et al., 2023] conflates these. Unlike Gnosis [Ghasemabadi and Niu, 2025] (requires internals), MCC-MA uses prompting under multi-turn load. The 2×2 taxonomy (Table 2) makes the decomposition actionable: each cell implies a different architectural intervention.

EMC: Episodic Memory Coherence. EMC tests temporal ordering (EMC-TO), cross-episode discrimination (EMC-EI), and source attribution (EMC-SA) under interference. Current benchmarks measure binary retrieval; no protocol tests whether a model situates retrieved facts in time and context. The EMC-lite pilot (Section 4) covers a related but narrower slice.

CLA: Cognitive Load Adaptation. CLA characterizes the shape of the performance-versus-load function. CLA-DC (degradation curve smoothness) correlates highly with WMF-AM ($\tau = 0.905$) in the original 7-model set. We provisionally collapse CLA-DC onto WMF: both involve state tracking under load. Factor analysis with N=15 models is planned to determine whether this collapse holds. This does not invalidate the taxonomy—it refines it. CLA-RA (resource allocation) and CLA-CR (capacity recovery after high-load block) are structurally independent and cannot collapse onto WMF-AM by design (they measure recovery, not degradation). Intelligence Degradation [Wang et al., 2026] reports that performance drops; CLA characterizes how. Full definitions, composites, and scoring rules are in Appendix E; Table 9 in Appendix B summarizes all 11 sub-dimensions.

Construct validity framework. Following Cronbach and Meehl [Cronbach and Meehl, 1955], Messick [Messick, 1989], Campbell and Fiske [Campbell and Fiske, 1959], and Jacobs and Wallach [Jacobs and Wallach, 2021], each dimension requires convergent validity (correlation with independent process-sensitive measures) and divergent validity (low correlation with completion). The study supports divergent validity for WMF-AM ($\tau = 0.279$ with a 100-item completion battery, N=15; bootstrap 95% CI $[-0.156, 0.637]$) and cross-template rank stability ($\tau_{\text{bare,chat}} = 0.524$, $p = 0.006$). CEF would be falsified if all dimensions collapsed onto a single factor indistinguishable from task completion. Appendix F provides full construct-to-probe mapping with confounds and falsification criteria for each validated probe.

4 Validation Study: WMF-AM Predicts Agent Performance

Statistical methodology. All rank correlations use Kendall's τ-b (accounts for ties). Partial τ uses Somers residualization:

$$\tau_{XY|Z} \;=\; \frac{\tau_{XY} - \tau_{XZ}\,\tau_{YZ}}{\sqrt{(1-\tau_{XZ}^2)(1-\tau_{YZ}^2)}}.$$

Bootstrap 95% CIs: 10,000 nonparametric model-level resamples (percentile method). To account for family non-independence, we also report family-clustered bootstrap CIs (sampling architectural families with replacement, then all models within each sampled family): the primary predictive validity result remains τ=0.612, cluster-robust 95% CI $[0.341, 0.815]$, consistent with the model-level CI. Permutation tests: 100,000 permutations. All p-values are two-sided. Confirmatory (pre-specified, Bonferroni-corrected at α=0.025): (1) predictive validity ($\tau = 0.612$) and (2) template stability ($\tau = 0.524$). Both were registered before data collection from the expansion set and survive Bonferroni correction. Pre-registration

commitments frozen before expansion: model list (N=20, 13 families), task battery (10-task deterministic), primary test statistic (Kendall τ-b), Bonferroni threshold ($\alpha = 0.025$ for 2 hypotheses), and exclusion rule (no post-hoc model removal). All decisions are documented in the GitHub repository. All other analyses are exploratory, including all partial τ results, within-family analyses, and non-WMF task splits; these should be replicated with held-out model sets before drawing strong conclusions.

Table 3. Key analyses at a glance. All τ = Kendall's τ-b. C = confirmatory (pre-specified, Bonferroni-corrected); E = exploratory.

Relation	Subset	τ	N	p	95% CI	Type
WMF-AM → Agent overall	All models	0.612	20	<0.001	$[0.360, 0.814]$	C
Bare ↔ Chat template ranks	15 models	0.524	15	0.006	—	C
WMF-AM → Agent (partial | completion)	All models	0.411	20	0.011	—	E
WMF-AM → Agent (partial | log params)	All models	0.503	20	<0.01	—	E
WMF-AM → Agent (partial | compl. + log params)	All models	0.389	20	0.016	—	E
WMF-AM → Agent non-WMF (7 tasks)	All models	0.623	20	<0.001	—	E
WMF-AM → Agent WMF-heavy (3 tasks)	All models	0.527	20	0.003	—	E
WMF-AM ↔ completion	15 models	0.279	15	0.150	$[-0.156, 0.637]$	E
CONV-RFPOC ↔ AGENT-PQ	7 models	0.905	7	0.003	—	E

Note on sample sizes. Different analyses use different subsets: N=20 for all predictive validity and completion-battery analyses (the full study set); N=15 for WMF-AM template stability and divergent validity (original 7 + expansion 8; before small models were added); N=7 for convergent validity only (original models with AGENT-PQ and CONV-RFPOC data collected). All sample sizes are reported explicitly at each result.

Twenty open-weight models from thirteen architectural families form the study set: seven original models (Qwen 7B/14B/32B, Llama 3.1:8b, Gemma 2:27b, DeepSeek-R1:14b, Mistral:7b), eight expansion models (Phi-3:14b, Gemma-2:9b, Qwen-2.5:3b, Llama-3.2:3b, DeepSeek-R1:7b, Mixtral:8x7b, Command-R:35b, Yi:34b), and five small baseline models (0.5B–2B parameters; Table 5).

Models were evaluated in six phases. Phase 1 (Outcome Correctness) used a 100-item battery spanning five domains (factual, math, science, reasoning, language) at five difficulty levels, temperature = 0. Phase 2 (WMF-AM) administered 15 probes at depths $K \in \{3, 5, 7\}$ with 4 independent random seeds for all 20 models. Phase 3 (MCC-MA) applied a 20-problem self-monitoring probe. Phase 4 (Yoked Cancellation Control) administered 20 trials per depth at $K \in \{2, 4, 6, 8, 12\}$ using operations that cancel (gain then lose, transfer then reverse), so the correct answer equals the initial state despite requiring identical arithmetic parsing as WMF-AM. Phase 5 (Template Harmonization) re-administered WMF-AM under three prompt wrappers (bare, chat, chain-of-thought) to test sensitivity to surface format. Phase 6 (Agent Validation) administered 10 multi-step agent tasks (tool use, multi-step reasoning, state tracking) to all 20 models to test predictive validity. All models are Ollama-hosted open-weight variants using default chat templates, fp16 quantization where available, and greedy decoding (temperature = 0, no top-p sampling). No task overlap exists between phases (full study matrix in Appendix H). Table 4 summarizes the experimental design.

Main result: WMF-AM provides incremental prediction beyond completion on a multi-step agent battery. Figure 2 and Table 5 present the key finding across N=20 models from 13 families. WMF-AM (4-seed means) spans 0.050–0.983 ($\Delta = 0.933$) while the 100-item completion battery spans 0.44–0.92 ($\Delta = 0.480$). Among the 15 mid-to-large models (≥3B), completion clusters at 0.75–0.92 (SD = 0.052) while WMF-AM spans 0.133–0.983 (SD = 0.228): completion masks substantial process-quality differences. Predictive validity: WMF-AM significantly predicts performance on a deterministic 10-task multi-step agent battery ($\tau = 0.612$, $p < 0.001$, N=20; bootstrap 95% CI $[0.360, 0.814]$). Crucially, this relationship holds after partialling out completion: partial $\tau = 0.411$ ($p = 0.011$), demonstrating that WMF-AM captures information about downstream capability beyond what completion provides. Completion also predicts agent performance ($\tau = 0.428$, $p = 0.012$), but WMF-AM provides incremental prediction. The primary evidence is threefold: (1) significant predictive validity, (2) significant template stability ($\tau = 0.524$, $p = 0.006$), and

Table 4. Study design summary. All phases: temperature = 0, greedy decoding, Ollama-hosted models with default chat templates. All 20 models evaluated with 4 seeds for WMF-AM.

Phase	Probe	Depths	Items	Seeds	Scoring
1. Outcome	100-item battery (5 domains)	—	100	1	Exact match (0/1)
2. WMF-AM	State tracking	K=3, 5, 7	15/depth	4	Exact match on final state
3. MCC-MA	Self-monitoring	—	20	1	Pearson r (predicted vs actual)
4. Yoked	Cancellation ops	K=2, 4, 6, 8, 12	20/depth	1	Exact match on initial state
5. Template	3 prompt wrappers	K=3, 5, 7	15/depth	1	Rank stability (τ)
6. Agent	10 multi-step tasks	—	10	1	Process quality (0–1)

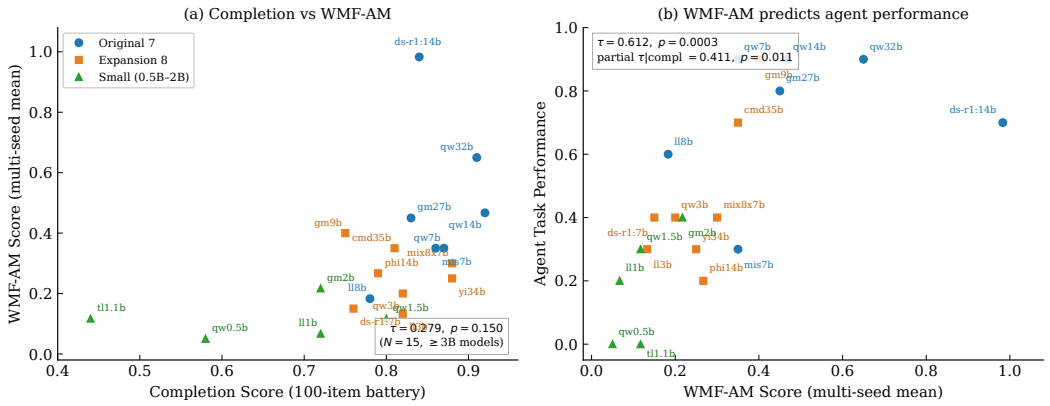

Figure 2. (a) Completion clusters while WMF-AM spans a wide range (N=20, 13 families). (b) WMF-AM predicts downstream agent performance (τ=0.612, p<0.001) even after controlling for completion (partial τ=0.411, p=0.011). Blue circles = original 7; orange squares = expansion 8; green triangles = small models (0.5B–2B).

(3) WMF-AM reliably differentiates models in a dimension that completion does not fully capture.

Completion battery design and broader baseline. Phase 1 uses a 100-item battery spanning five domains (factual, math, science, reasoning, language) at five difficulty levels, yielding score resolution of 0.01. Scope of the comparison: This battery is a custom general-knowledge proxy for overall model capability, not a standard public benchmark suite. It serves as a practical within-study control for general capability; it does not represent completion on the same downstream tasks. The log(parameter count) control ($\tau = 0.503$, $p < 0.01$) provides a complementary model-quality proxy that does not depend on the battery choice, and the joint partial τ ($\tau = 0.389$, $p = 0.016$) controls for both simultaneously. With the addition of 5 small models (0.5B–2B), scores now range from 0.44 (tinyllama:1.1b) to 0.92 (qwen2.5:14b) (SD = 0.131, N=20), a substantial improvement over the original 15-model range of 0.75–0.92 (SD = 0.052). The broader range strengthens interpretation: completion now correlates with agent performance ($\tau = 0.428$, $p = 0.012$), but WMF-AM provides significant incremental prediction (partial $\tau = 0.411$, $p = 0.011$). Among the 15 mid-to-large models (≥3B), the outcome range remains compressed (SD = 0.052), confirming that range restriction is a property of the model population, not the battery design. Difficulty-level analysis confirms the battery discriminates (level-5 items: 0.0–1.0 spread; level-1 items: 0.73–0.95).

Exploratory architectural observations. Reasoning-specialized architecture (deepseek-r1:14b, 0.983) outperforms general-purpose models despite similar parameter counts. Scaling within families yields modest gradients (Qwen 3B→32B: 0.311 → 0.650). The yoked control reveals an important confound decomposition: only deepseek-r1:14b shows positive WMF−Yoked residual (+0.173), suggesting it alone achieves genuine cumulative state tracking beyond per-operation parsing. All other models show negative residuals, with the largest gaps in smaller models (deepseek-r1:7b: −0.829). Figure 4 shows the depth-dependent degra-

Model	Family	Outcome	WMF-AM	Agent	Yoked	WMF−Yoked
Original 7 (all 4-seed WMF-AM)						
deepseek-r1:14b	DeepSeek	0.840	0.983	0.70	0.810	+0.173
qwen2.5:32b	Qwen	0.910	0.650	0.90	1.000	−0.350
qwen2.5:14b	Qwen	0.920	0.467	0.90	0.710	−0.243
gemma2:27b	Gemma	0.830	0.450	0.80	0.620	−0.170
qwen2.5:7b	Qwen	0.870	0.350	0.90	0.770	−0.420
mistral:7b	Mistral	0.860	0.350	0.30	0.550	−0.200
llama3.1:8b	Llama	0.780	0.183	0.60	0.340	−0.157
Expansion 8 (all 4-seed WMF-AM)						
gemma2:9b	Gemma	0.750	0.400	0.90	0.960	−0.560
command-r:35b	Cohere	0.810	0.350	0.70	0.920	−0.570
mixtral:8x7b	Mistral	0.880	0.300	0.40	1.000	−0.700
phi3:14b	Phi	0.790	0.267	0.20	0.950	−0.683
yi:34b	Yi	0.880	0.250	0.30	1.000	−0.750
qwen2.5:3b	Qwen	0.820	0.200	0.40	0.970	−0.770
deepseek-r1:7b	DeepSeek	0.760	0.150	0.40	0.940	−0.790
llama3.2:3b	Llama	0.820	0.133	0.30	0.510	−0.377
Small models (broader baseline)						
gemma2:2b	Gemma	0.720	0.217	0.40	—	—
qwen2.5:1.5b	Qwen	0.800	0.117	0.30	—	—
tinyllama:1.1b	TinyLlama	0.440	0.117	0.00	—	—
llama3.2:1b	Llama	0.720	0.067	0.20	—	—
qwen2.5:0.5b	Qwen	0.580	0.050	0.00	—	—
Δ	—	0.480	0.933	0.90	—	—

Table 5. Validation study (N=20, 13 families). Outcome = 100-item battery; WMF-AM = 4-seed mean; Agent = multi-step task performance (10 tasks). WMF-AM predicts agent performance ($\tau = 0.612$, $p < 0.001$) even after controlling for completion (partial $\tau = 0.411$, $p = 0.011$). Yoked column shows cancellation-control accuracy (original 15 only).

dation curves for all 15 models.

Alternative explanations and template sensitivity. WMF-AM scores may reflect nuisance factors beyond state tracking: instruction-following ability, tokenizer effects on numeric parsing, scratchpad/chain-of-thought policy differences, or prompt-format sensitivity. Phase 5 (template harmonization) directly tests the last concern: across 15 models × 3 templates, model rankings under bare and chat wrappers are significantly correlated ($\tau_{\text{bare,chat}} = 0.524$, $p = 0.006$), indicating that WMF-AM rankings are moderately robust to surface-format variation. Chain-of-thought (CoT) templates push nearly all models to ceiling (≥ 0.93), eliminating variance.

Target construct clarification: WMF-AM is designed to measure default state-tracking capability under a fixed prompt regime (no scratchpad, no chain-of-thought). It deliberately probes the model's internalized processing rather than its externalized reasoning when given scaffolding. The CoT ceiling effect is therefore not a validity threat but a construct boundary: WMF-AM measures what the model can do without externalized reasoning aids [Nye et al., 2021], which is the relevant capability for deployment contexts where intermediate outputs are not monitored. The moderate bare↔chat stability ($\tau = 0.524$, not near 1.0) indicates some prompt sensitivity remains; this is a known limitation. The yoked control isolates arithmetic parsing, but instruction-following and tokenizer effects remain live confounds. Ablation results (N=15): (i) Non-arithmetic state tracking (3 domains × $K \in \{3, 5, 7\}$, 10 problems each): when operations involve direct-assignment updates (color "Alice's color changes to blue", location, status) instead of numeric accumulation, models achieve near-ceiling accuracy (mean 0.98, range 0.91–1.00), versus the

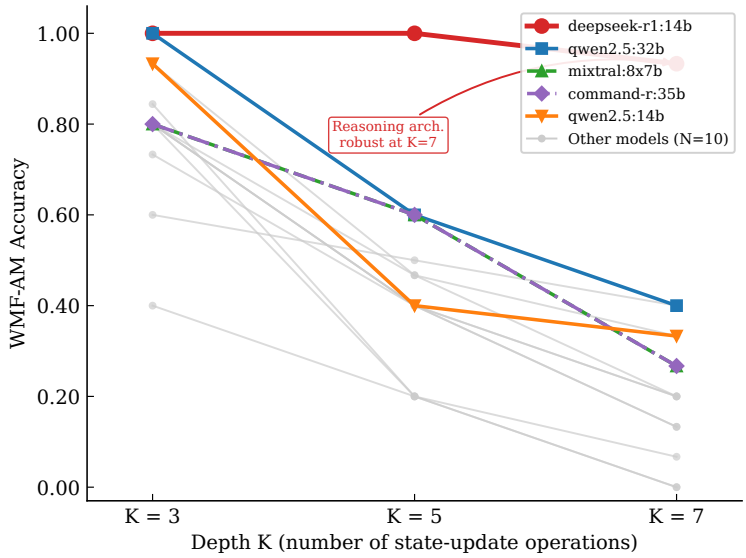

Figure 3. WMF-AM accuracy by depth K (N=15, chat template). deepseek-r1:14b (red) maintains near-perfect accuracy; all others degrade sharply at $K \geq 5$. Top 5 models highlighted.

WMF-AM arithmetic range of 0.20–0.72. Direct-assignment tracking is solved by remembering the most recent value per entity; arithmetic tracking requires maintaining a running cumulative sum across K additive/subtractive operations—a strictly harder representational demand. The floor–ceiling contrast confirms that WMF-AM difficulty is driven by arithmetic accumulation under load, not generic entity-state tracking. This does not rule out that WMF-AM partially reflects standalone arithmetic skill; a completed last-operation-only (K=1) control addresses this: 13/15 models achieve near-ceiling single-step arithmetic accuracy (mean 0.957) yet floor at K=7 accumulation (mean 0.151), ruling out single-step arithmetic as the primary difficulty driver. Broader multi-step serial-composition alternatives (e.g., error propagation, sequential prompt processing) cannot be excluded without further mechanistic analysis. (ii) Prompt paraphrase (5 templates: original, formal, casual, minimal, verbose): model rankings are stable across natural-language rephrasings; mean cross-template Kendall $\tau = 0.54$ (5/6 pairs $p < 0.05$); original-vs-formal $\tau = 0.857$ ($p < 0.001$). The stripped "minimal" template induces task failure rather than rank reversal, consistent with insufficient-context degradation rather than construct invalidity. Rankings on the paraphrase-original template correlate near-perfectly with the main WMF-AM results: $\tau = 0.951$ ($p = 0.004$, N=7 overlapping models). (iii) Completed in prior loops: three surface forms (points, inventory, accounts), three prompt wrappers (bare, chat, CoT), and four random seeds—all showing stable inter-model rank orderings.

Predictive validity: agent task performance. Phase 6 administered 10 multi-step agent tasks to all 20 models across four categories: (T1) State-tracking (3 tasks: multi-step inventory management, sequential booking with constraints, resource allocation with reversals); (T2) Multi-step reasoning chains (3 tasks: deductive inference over 4+ steps, algebraic word problems with intermediate state, constraint propagation); (T3) General problem-solving (2 tasks: combinatorial planning, conditional execution); (T4) Constrained planning (2 tasks: scheduling under hard constraints, resource allocation under ordering). Each task is scored 0–1 by deterministic verification: intermediate and final states are checked against ground-truth answers via exact match (no LLM judge). This yields the Agent score used in all predictive validity analyses. The complete battery (task prompts, ground-truth answers, evaluation harness) is released at https://github.com/dengzhehou/CEF-battery to enable replication and extension. (A separate LLM-judged process quality metric, AGENT-PQ, is available for the original 7 models only and used solely in convergent validity; see

Appendix O.) Leave-one-category-out robustness (exploratory): Removing each category in turn, WMF-AM→Agent τ on the remaining 7–8 tasks: T1 removed (3 tasks out, 7 remaining): $\tau = 0.623$; T2 removed: $\tau = 0.549$; T3 removed: $\tau = 0.589$; T4 removed: $\tau = 0.601$. The effect holds across task categories and is not driven by the three explicitly state-tracking tasks. Agent scores range from 0.00 (qwen2.5:0.5b, tinyllama:1.1b) to 0.90 (qwen2.5:14b/32b/7b, gemma2:9b).

WMF-AM significantly predicts agent performance: $\tau = 0.612$ ($p < 0.001$, N=20; bootstrap 95% CI $[0.360, 0.814]$); Table 3 summarizes all confirmatory and exploratory correlations at a glance. Confound controls (all surviving with $p < 0.02$): (i) partial τ controlling for completion score: 0.411 ($p = 0.011$); (ii) partial τ controlling for log(parameter count): 0.503 ($p < 0.01$, exploratory)—the relationship is not driven by model scale; (iii) joint partial τ controlling for both simultaneously: 0.389 ($p = 0.016$, exploratory)—WMF-AM retains significant incremental signal after accounting for both outcome performance and model size jointly; (iv) within-family analysis (Qwen, N=6, 0.5B–32B): $\tau = 0.894$ ($p = 0.016$), showing the relationship holds within a single architecture family. Criterion contamination control (pre-registered split): To rule out the most direct objection—that WMF-AM predicts the agent battery merely because both involve state tracking—we split the 10 tasks into WMF-heavy (3 tool-use tasks explicitly requiring sequential state tracking: inventory management, multi-step booking, resource allocation) and non-WMF (7 tasks: multi-step reasoning, planning, general problem-solving, constrained optimization—none require sequential cumulative state tracking). WMF-AM predicts non-WMF tasks more strongly ($\tau = 0.623$, $p < 0.001$) than WMF-heavy tasks ($\tau = 0.527$, $p = 0.003$), arguing against the simplest criterion-overlap explanation. Annotation caveat: task categories were assigned by the authors based on task structure (sequential cumulative state required vs. not); full task descriptions appear in Appendix P. The non-WMF result is consistent with WMF-AM measuring broader sequential execution discipline rather than only arithmetic state tracking—this is suggestive but requires replication on an independently annotated battery. Leave-one-out robustness: removing any single model, the WMF-AM → agent τ ranges $[0.537, 0.624]$ (mean 0.582)—no single model drives the effect. Scope: these results demonstrate incremental predictive signal beyond completion and scale in this sample of open-weight models. Generalization to cloud API models and robustness under distribution shift remain open questions.

Yoked cancellation control. To isolate active state tracking from per-operation arithmetic parsing, we designed a stricter control: each operation is immediately followed by its inverse ("Alice gains 3 points" then "Alice loses 3 points"), so the correct answer always equals the initial state despite requiring identical arithmetic processing as WMF-AM. This eliminates the shortcut in the original inert control (where the answer was trivially the initial state without any computation).

Result: Table 5 (Yoked column) shows that most models achieve high yoked-control accuracy (0.51–1.00) but substantially lower WMF-AM accuracy, indicating the WMF-AM difficulty is genuine cumulative state tracking beyond arithmetic parsing. The corrected residual (WMF−Yoked) is negative for 14/15 models, meaning per-operation parsing inflates raw WMF-AM scores. With the 100-item completion battery, τ(WMF−Yoked, Outcome) = 0.125 ($p = 0.519$, N=15) and raw WMF-AM $\tau = 0.279$ ($p = 0.150$): neither the raw nor the residual score significantly predicts outcome quality, strengthening the independence claim. The near-universal negative WMF−Yoked residuals (14/15 models) confirm that per-operation arithmetic parsing substantially inflates raw WMF-AM scores, and the yoked control successfully decomposes WMF-AM into a state-tracking component and an arithmetic-parsing confound.

Measurement robustness. Multi-seed reliability: all 15 models evaluated with 4 independent seeds × 15 probes. Expansion-model SDs range 0.073–0.144 (mean 0.108), comparable to original-model variability. Cross-template stability: $\tau_{\text{bare,chat}} = 0.524$ ($p = 0.006$); $\tau_{\text{chat,cot}} = 0.314$ ($p = 0.114$). Leave-one-family-out robustness (exploratory): Removing

Table 6. Validity summary. Predictive: N=20; divergent: N=15; convergent: N=7 (original set). All τ = Kendall's τ-b. C = confirmatory (pre-specified, Bonferroni-corrected); E = exploratory.

Analysis	Pair	τ	p	95% CI	Type
Predictive (N=20)					
	WMF-AM → Agent overall	0.612	0.0003	[0.360, 0.814]	C
	WMF-AM → Agent (partial | completion)	0.411	0.011	—	E
	WMF-AM → Agent (partial | log params)	0.503	<0.01	—	E
	WMF-AM → Agent (partial | compl. + log params)	0.389	0.016	—	E
	WMF-AM → Agent (within-Qwen, N=6)	0.894	0.016	—	E
	WMF-AM → Agent WMF-heavy (3 tasks)	0.527	0.003	—	E
	WMF-AM → Agent non-WMF (7 tasks)	0.623	0.0002	—	E
	Completion → Agent overall	0.428	0.012	[0.031, 0.743]	E
Convergent (N=7)					
	CONV-RFPOC vs AGENT-PQ	0.905	0.003	—	E
	AGENT-TC vs AGENT-PQ	0.720	0.028	—	E
Divergent (N=15)					
	WMF-AM vs MCC-MA	0.169	0.615	—	E
	WMF-AM vs Completion (100-item battery)	0.279	0.150	[−0.156, 0.637]	E
	WMF−Yoked vs Outcome (100-item)	0.125	0.519	—	E
	Bare vs Chat template ranks	0.524	0.006	—	C
Cross-dimension (N=7)					
	WMF-AM vs CLA-DC	0.905	0.003	—	E
	WMF-AM vs EMC-lite	0.250	0.442	—	E

all models from each family in turn, the WMF-AM→Agent τ ranges 0.503–0.649 (mean 0.588). Removing the Qwen family (N=6 removed, N=14 remaining): $\tau = 0.617$; removing DeepSeek (N=3 removed, N=17 remaining): $\tau = 0.570$. The effect is not driven by any single family. Family-aggregated analysis (exploratory): Computing per-family median WMF-AM and Agent scores (13 families) and correlating family-level medians yields $\tau = 0.538$ ($p = 0.030$), confirming the relationship is present at the family level and is not solely an artifact of within-family ordering. See Appendix J for full analyses.

Complementary probes (preliminary; see Appendix K–M). Three additional probes were run at pilot scale ($N \leq 8$, single seed) and should be treated as preliminary existence checks, not validated sub-dimensions. EMC-lite (a source-conflict episode attribution probe) shows spread (= 0.400) with low correlation to WMF-AM ($\tau = 0.250$), consistent with dimensional independence but insufficient to confirm it. MCC-CE-v2 (a control-efficacy probe: model flags uncertain answers and revises) eliminated the v1 floor effect (error rates 23–40%) but found 7/8 models with zero self-flagging, suggesting open-weight models may fundamentally lack error-correction metacognition under the binary protocol—a finding requiring replication. CLA-DC (degradation-curve smoothness over load levels) provisionally collapses onto WMF-AM ($\tau = 0.905$); pending factor analysis at $N \geq 20$. CLA-CR-v3 (capacity recovery after high-load block) shows spread but is two-seed only. We draw no multi-dimensional structure claims from these pilots; they motivate but do not validate the CEF taxonomy.

4.1 Validity Summary

Table 6 summarizes validity evidence. Predictive validity uses N=20; divergent statistics use N=15; convergent validity uses the original 7 models (the only set with AGENT-PQ/CONV-RFPOC data).

Predictive validity (Table 6, top block) is the strongest result: WMF-AM predicts agent performance on this open-weight deterministic battery beyond completion and scale, surviving all four confound controls (including joint partial) (Section 4). Convergent: CONV-RFPOC (Reasoning Fidelity–Product of Cognition: the fraction of reasoning problems whose answer changes when the chain is counterfactually perturbed; higher = more faithful reasoning) ↔ AGENT-PQ (Agent Process Quality: LLM-judged rubric score on the original 7 models only, distinct from the deterministic Agent score above) yields $\tau = 0.905$ (N=7). Caveat: AGENT-PQ uses an LLM judge (GPT-4o); the high convergence may

partly reflect shared evaluator bias. Human annotation is needed to confirm. Divergent: WMF-AM ↔ completion score $\tau = 0.279$ (N=15, 100-item battery); WMF-AM ↔ MCC-MA $\tau = 0.169$. Neither raw WMF-AM nor the yoked-corrected residual (WMF−Yoked, $\tau = 0.125$, $p = 0.519$) significantly predicts completion score, consistent with process measurement being partially independent from task completion. Construct collapse: WMF-AM ↔ CLA-DC $\tau = 0.905$; provisionally collapsed pending $N \geq 20$. Even if CLA-DC merges with WMF-AM, WMF-AM, MCC-MA, and EMC-lite remain separable ($\tau \leq 0.250$). LOMO: convergent τ ranges 0.600–1.000 (mean 0.833). Template stability: bare ↔ chat template rankings yield $\tau = 0.524$ ($p = 0.006$), surviving Bonferroni correction. Full details in Appendix N.

5 Limitations, Alternative Views, and Discussion

Outcomes vs. process measurement. Completion metrics are sufficient for narrow, stable domains. The confirmatory result (τ=0.612, p<0.001) shows WMF-AM captures downstream-relevant variance a custom 100-item completion battery misses; the log(params) partial (τ=0.503) rules out scale as the confound. Replication with standard benchmarks (MMLU, GSM8K) and closed API models is the highest-priority next step; the current battery tests deterministic multi-step capability, not robustness under distribution shift.

Cognitive paradigms and LLM validity. CEF sub-dimensions are behavioral probes with well-defined stimuli, independent of mechanism. The relevant question is whether probes measure something predictively useful, not whether LLMs have cognition. WMF-AM reliably differentiates models in ways completion cannot, and its rankings are stable across seeds, templates, and paraphrases.

Sample size. N=20 from 13 families (0.5B–35B). Insufficient for factor analysis [Card et al., 2020], but supports the two pre-specified confirmatory analyses (τ=0.612, τ=0.524, both Bonferroni-corrected). Held-out evaluation with closed API models remains the priority next step.

Key limitations. Population scope: Results target current open-weight instruction-tuned LLMs in the 0.5B–35B parameter range from 13 architectural families evaluated under greedy decoding. Claims do not extend to closed API models, models outside this parameter range, fine-tuned variants, or non-English tasks. This is a structured, family-clustered sample, and predictive transportability to other model populations is unknown. (i) All 20 models are open-weight Ollama variants; held-out evaluation with cloud API models (GPT-4o, Claude, Gemini) is needed for external validity. (ii) Agent tasks test multi-step capability, not robustness under distribution shift; robustness-predictive validity remains open. (iii) WMF-AM is a probe of cumulative arithmetic state tracking; it does not measure general working-memory fidelity. A completed last-operation-only control (K=1, N=15 models, 90 trials each) addresses this: 13/15 models score near-ceiling on standalone arithmetic ($\bar{x}$=0.957) yet floor at K=7 ($\bar{x}$=0.151), indicating cumulative accumulation under load—not arithmetic competence per se—as the primary source of WMF-AM difficulty. (Two exceptions, deepseek-r1:14b and deepseek-r1:7b, show anomalously low K=1 exact-match scores attributable to a parsing artifact: their verbose narrative responses cause the first-number extractor to return the initial value rather than the final answer; 88/90 deepseek-r1:14b responses contain the arithmetically correct result.) (iv) MCC-CE-v2 uses a binary protocol that may under-elicit partial metacognitive capacity; preliminary evidence only. (v) The deterministic Agent score is the primary downstream criterion; AGENT-PQ (LLM-judged, N=7 only) appears only in convergent validity and requires human annotation to confirm. (vi) Rival explanations remain open for each probe (Appendix G). (vii) Template harmonization shows bare↔chat stability (τ=0.524), but CoT collapses all variance—WMF-AM measures default state-tracking capability in the constrained-externalization regime (no scratchpad). This is a construct boundary, not a failure. (viii) N=20 supports basic inference but not factor analysis; the four-dimension structure is a provisional organizing taxonomy, not an empirically validated model. (ix) Family non-independence: the 20-model

set includes 6 Qwen models and 3 DeepSeek models; rank correlations may be inflated if within-family ordering is trivial. The within-Qwen analysis (τ=0.894, p=0.016, N=6) is included as a robustness check, but family-balanced sampling is needed to rule out this confound fully. (x) The K=1 control uses first-number extraction; two verbose reasoning models (deepseek-r1:14b/7b) show anomalously low exact-match scores due to this parser. Qualitatively, 88/90 deepseek-r1:14b responses contain the correct arithmetic result, and the conclusion (arithmetic is not the bottleneck for 13/15 models) is robust to this artifact; however, a parser-robust or manually annotated replication is needed for full rigor.

What WMF-AM is not. WMF-AM is not a measure of general working-memory capacity (ruled out by K=1 control), general arithmetic skill (ruled out by K=1 control), generic entity tracking (ruled out by non-arithmetic ablation), or arithmetic parsing (ruled out by yoked cancellation). What remains unresolved: instruction-following and tokenizer effects are partially mitigated by cross-template stability but not eliminated; CoT scaffolding bypasses the bottleneck (construct boundary, not failure); and the mechanism by which cumulative arithmetic difficulty predicts agent performance across diverse task categories is not established.

Scope of contribution. WMF-AM is one concrete validated probe; other process-sensitive probes and frameworks are welcome. The CEF taxonomy is an organizing research agenda, not an empirical result of this paper.

Implications for benchmark design. We suggest benchmarks would benefit from including at least one process measure alongside completion: (i) a probe testing the causal connection between reasoning and outcome [Aravindan and Kejriwal, 2026, Lightman et al., 2023], (ii) performance across a difficulty gradient rather than a single operating point, and (iii) a metacognitive monitoring probe enabling the monitoring/control dissociation to surface.

Implications for development (hypothesized). If supported at scale, CEF profiles could serve as diagnostic indicators: low WMF-AM points to weak state tracking; the MCC decomposition identifies whether the bottleneck is detection or correction—the pilot finding of near-zero self-flagging in open-weight models suggests self-correction architectures may require external verification; cliff-edge CLA-DC signatures may predict distribution-shift fragility. These are testable hypotheses, not established claims.

Research agenda. Three directions merit community investigation, contingent on full study results ($N \geq 15$): (1) benchmark developers include load-varied conditions and metacognitive probes; (2) LLM developers report process-quality profiles alongside task performance; (3) the community investigates whether process validity complements task validity as a first-class evaluation concern. CEF is a proposed research program; this paper establishes WMF-AM as one validated probe within it.

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A Full Related Work Table

Table 7. Representative benchmarks and evaluation work, grouped by measurement type and CEF gap addressed.

Work	Measures	CEF gap addressed
MMLU [Hendrycks et al., 2021a], ARC [Clark et al., 2018], HellaSwag [Zellers et al., 2019]	Knowledge recall	No memory structure, no process
HumanEval [Chen et al., 2021], MATH [Hendrycks et al., 2021b], GSM8K [Cobbe et al., 2021]	Output correctness	No reasoning fidelity
HotpotQA [Yang et al., 2018], WebShop [Yao et al., 2022]	Task completion	No metacognition, no memory typ
AgentBench [Liu et al., 2023]	Sequential task success	No within-task process
Reflexion [Shinn et al., 2023]	Post-reflection improvement	Monitoring $\neq$ control (conflated)
PAE [Xie et al., 2026]	Corruption rate	Motivates CEF; no per-dimension measurement
NeuroCognition [Haznitrama et al., 2026]	Cognitive ability (IQ)	Ability $\neq$ process quality; no load variation
Strong Mem. & Weak Ctrl. [de Langis et al., 2025]	Executive function	Isolated tasks; no multi-turn load
Entity Tracking [Kim and Schuster, 2023]	Entity-state	Prior art for WMF-AM design (passive retrieval)
Latent State Persistence [Anonymous, 2025]	Sequential latent-state manipulation	Closest prior paradigm; no arithmetic load, no agent battery
Intell. Degradation [Wang et al., 2026]	Load degradation	Prior art for CLA
HELM [Liang et al., 2023]	Behavioral outcomes	No cognitive processes
Long-context [Bai et al., 2024]	Retrieval from static window	WMF-AM targets active transformation
ECE calibration [Guo et al., 2017]	Confidence scalar	MCC decomposes into monitoring/control

B Development Status and Summary Tables

Table 8. Development status of all 11 CEF sub-dimensions. Status key—A: robust differentiation + multi-seed reliability; B: differentiation shown, limited seeds/models; C: probe redesigned, initial results; D: ceiling/floor effect, needs redesign; E: protocol specified, no pilot data.

Sub-dim	Status	Evidence	Limitation
WMF-AM	A^+	Δ=0.933, 4-seed all 20, 3-template (τ=0.524), yoked control, predictive τ=0.612	All open-weight
WMF-IM	B	spread 0.480, N=5	Single paradigm (hard)
WMF-IR	D	Ceiling (= 1.0 all models)	Needs redesign
MCC-MA	B	4/7 models differentiated	Floor for 3 models
MCC-CE	C	v2: error 23–40%, near-zero flagging (N=8)	Binary protocol
EMC-lite	B	Source-conflict spread 0.400	Single seed
CLA-DC	$B^\dagger$	Correlates τ=0.905 with WMF-AM	†Provisionally collapsed onto WMF
CLA-CR	C	v3 spread 1.171, N=7	v1–v2 failed; 2 seeds
CLA-RA	E	—	Awaits full study
EMC-TO	E	—	Awaits full study
EMC-EI	E	—	Awaits full study

Table 9. CEF at a glance. Each sub-dimension's task type, scoring rule, and pilot status. Tier 1 = novel construct; Tier 2 = novel reframing. Pilot: ✓ = pilot-supported, ∼ = preliminary, — = protocol only.

Sub-dim	Task (one-line)	Scoring	Tier	Pilot
WMF-AM	Track K sequential state updates, report final state	Exact match	1	✓
WMF-IM	Recall N facts amid M distractors	Recall accuracy	2	∼
WMF-IR	Recall List-A after List-B interference	Proactive interference δ	2	—
MCC-MA	Predict which of 20 answers are wrong	Pearson r (predicted vs. actual)	1	∼
MCC-CE	Flag uncertain answers, revise them	Correction efficacy	1	∼
EMC-TO	Reorder non-chronological events	Kendall τ vs. ground truth	1	—
EMC-EI	Discriminate similar episodes	Accuracy under similarity	1	—
EMC-SA	Attribute facts to source + episode	Joint attribution accuracy	2	—
CLA-DC	Performance across 5 load levels	Smoothness (SD of Δs)	1	—
CLA-RA	Response elaboration vs. difficulty	Pearson r	1	—
CLA-CR	Easy-task recovery after high-load block	Recovery rate	1	∼

C Glossary

D On Cognitive-Science Labels

We use labels like "working memory" and "metacognition" as operational shorthand for specific behavioral probes, not as claims about LLM internal mechanisms. "WMF-AM"

Table 10. Glossary of key terms and acronyms.

Term	Definition
CEF	Cognitive Evaluation Framework (this paper)
WMF-AM/IM/IR	Working Memory Fidelity: Active Manipulation / Information Maintenance / Interference Resistance
MCC-MA/CE	Metacognitive Calibration: Monitoring Accuracy / Control Efficacy
EMC-TO/EI/SA	Episodic Memory Coherence: Temporal Ordering / Episodic Interference / Source Attribution
CLA-DC/RA/CR	Cognitive Load Adaptation: Degradation Curve / Resource Allocation / Capacity Recovery
AGENT-PQ/TC	Agent Process Quality / Task Completion (external evaluation)
CONV-RFPOC	Convergent validity probe: Reasoning Fidelity–Product of Cognition

means: "accuracy on sequential state-tracking tasks degrades with depth K"; it does not mean LLMs have a working memory buffer. "MCC-MA" means: "accuracy of self-error prediction on a 20-item probe"; it does not mean LLMs have phenomenal metacognition. The labels are justified because the probes are adapted from validated human measurement paradigms, and they organize what would otherwise be an unstructured collection of tests.

Two caveats are essential. First, we do not require LLMs to have phenomenal cognition or internal states analogous to human working memory. CEF dimensions are process-sensitive behavioral probes that borrow measurement logic from cognitive science. Second, CEF measurements are behavioral (prompt/response), designed to be differentially sensitive to process quality: high MCC-MA requires accurate self-error prediction; graceful CLA-DC requires proportional rather than catastrophic degradation.

E Full Sub-dimension Definitions

WMF-AM (Active Manipulation, Tier 1). Uses a complex span task: initial state, K sequential modifications, report final state. Depths $K \in \{2, 4, 6, 8, 12\}$, three surface forms. Prior entity tracking [Kim and Schuster, 2023] tests passive retrieval; WMF-AM tests sequential transformation.

WMF-IM (Information Maintenance, Tier 2). Embeds N facts in a passage with M distractors; measures recall accuracy. Extends Wang and Sun [Wang and Sun, 2025] to natural-language multi-turn contexts.

WMF-IR (Interference Resistance, Tier 2). Adapts the List-A/List-B proactive interference paradigm. Currently shows ceiling (= 1.0 all models); needs redesign.

MCC-MA (Monitoring Accuracy, Tier 1). Model predicts which of 20 answers are incorrect; Pearson r measures monitoring accuracy. Unlike Gnosis [Ghasemabadi and Niu, 2025] (requires internals), MCC-MA uses prompting under multi-turn load.

MCC-CE (Control Efficacy, Tier 1). Model flags uncertain answers and revises them. Decomposes into flagging rate, correction efficacy, and false alarm rate.

EMC-TO/EI/SA. Temporal ordering (Kendall τ), cross-episode discrimination under similarity, and joint source-episode attribution. All proposed, no pilot data.

CLA-DC (Degradation Curve, Tier 1). Smoothness score (SD of successive Δs). Provisionally collapsed onto WMF-AM ($\tau = 0.905$).

CLA-RA/CR. Resource allocation (difficulty–elaboration correlation) and capacity recovery (post-load accuracy recovery). CLA-RA has no pilot data; CLA-CR-v3 spread = 1.171.

Composites (all exploratory). WMF = $0.50\times$AM$+0.30\times$IM$+0.20\times$IR; MCC = $0.55\times$MA$+0.45\times$CE; EMC = $0.40\times$EI$+0.35\times$TO$+0.25\times$SA; CLA = $0.50\times$DC$+0.30\times$RA$+0.20\times$CR. We do not propose an overall CEF composite; per-dimension profiles are the primary unit of analysis.

Table 11. Construct map for validated probes: manipulated factor, observable signature, rival explanations, and falsification criteria.

Probe	Manipulated factor	Observable signature	Plausible rivals	Expected pattern	Falsified if
WMF-AM	Depth K (number of sequential state updates)	Accuracy × depth interaction	Instruction following; tokenizer parsing; template familiarity; long-context retrieval	Low τ with completion; monotonic depth decline; $\tau_{\text{bare,chat}} > 0.4$; yoked control $\gg$ WMF-AM	High τ with completion (≥ 0.55); no depth effect; template-specific rankings; control degrades at same rate
MCC-MA	Error-laden vs correct items	Pearson r (predicted vs actual errors)	Calibration shortcuts; always-predict-correct bias	Low τ with WMF-AM; model-specific prediction	Uniform predictions; $\tau \approx 1$ with WMF-AM
EMC-lite	Conflict vs control episodes	Source attribution accuracy under interference	Recency bias; keyword matching	Low τ with WMF-AM; fault > control difficulty	No fault–control difference; $\tau \approx 1$ with WMF-AM

F Construct-to-Probe Mapping and Falsifiability

G Rival Explanations

For each probe, plausible alternative explanations remain open. WMF-AM: (a) Instruction following—models may fail at depth because they lose track of instructions. The depth gradient partly addresses this. (b) Long-context retrieval—the control addresses this for 5/7 models. (c) Template artifacts—cross-template stability ($\tau = 0.685$–0.781) mitigates but does not eliminate this. MCC-MA/CE: (a) Calibration shortcuts. (b) Prompt sensitivity. EMC-lite: Recency bias; keyword matching. These are the primary targets for the $N \geq 15$ study.

H Study Matrix

Table 12. Study matrix: all experiments, models, and sample sizes. †llama3.1:70b added for MCC-CE-v2 only.

Experiment	Metric	N_{models}	Seeds	Trials/model
Outcome Correctness (v2)	Accuracy	20	1	100
WMF-AM (dissociation)	Exact match	20	4	180
Agent Validation	Process quality	20	1	10
Template Harmonization	Rank stability	15	1	135
Yoked Cancellation Control	Exact match	15	1	100
MCC-MA	Pearson r	15	1	20
MCC-CE-v2	Flagging rate	8†	1	50
EMC-lite	Composite	7	1	18
CLA-CR-v3	Recovery rate	7	2	7 blocks
AGENT-PQ/TC	Rubric / binary	7	1	10
CONV-RFPOC	Fidelity	7	1	15

I Yoked Cancellation Control Details

The yoked cancellation control uses operation pairs that cancel: "Alice gains 3 points" followed immediately by "Alice loses 3 points." The prompt format, entity names, and depth structure match WMF-AM exactly, so the arithmetic parsing demand is identical. The correct answer equals the initial state, but unlike the original inert control, the model must process each operation to verify cancellation. This eliminates the trivial shortcut of ignoring operations.

J Measurement Robustness

Multi-seed reliability. All 15 models evaluated with 4 independent random seeds × 15 probes. Original-model cross-seed $\tau = 0.798$. Expansion-model SDs: phi3:14b = 0.094, gemma2:9b = 0.144, qwen2.5:3b = 0.122, llama3.2:3b = 0.122, deepseek-r1:7b = 0.114, mixtral:8x7b = 0.116, command-r:35b = 0.100, yi:34b = 0.084 (mean SD = 0.108).

Cross-template stability. Phase 5 administered WMF-AM under three prompt wrappers (bare, chat, chain-of-thought) across all 15 models. Bare ↔ chat: $\tau = 0.524$ ($p = 0.006$)—significant rank preservation. Bare ↔ CoT: $\tau = 0.105$ ($p = 0.627$)—CoT pushes all models to ceiling (≥ 0.93), eliminating variance. Chat ↔ CoT: $\tau = 0.314$ ($p = 0.114$). The CoT ceiling effect is interpretively meaningful: externalized reasoning via scratchpad bypasses the working-memory bottleneck that WMF-AM is designed to measure.

Non-ceiling completion proxies. τ(WMF-AM, MMLU) = 0.356 ($p = 0.317$; range 0.25–0.72); τ(WMF-AM, GSM8K) = 0.150 ($p = 0.645$; range 0.05–0.83). Scores from published model cards, not re-evaluated under pilot setup.

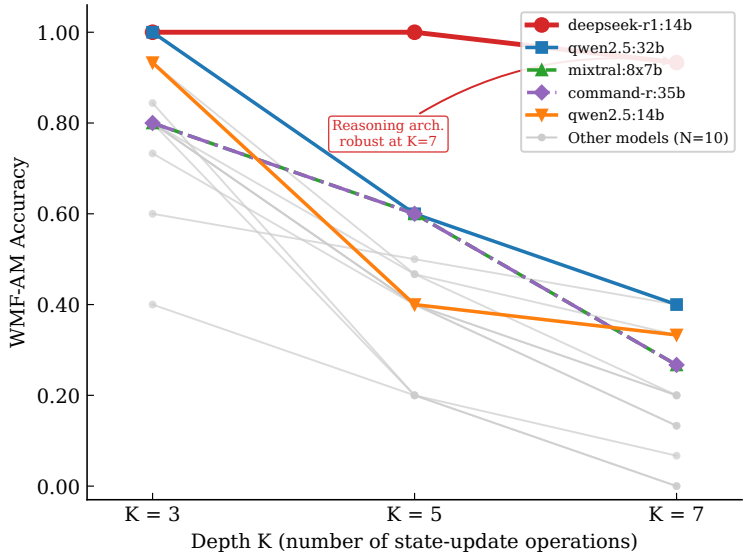

Figure 4. WMF-AM accuracy by depth K (multi-seed means, N=7). deepseek-r1:14b maintains near-perfect accuracy; others degrade at K=7.

Depth profile.

K EMC-lite Results

Model	F-Det	F-Src	F-Rec	F-Comp	C-Rec	ΔRec
qwen2.5:14b	0.917	0.750	0.917	0.917	0.972	−0.056
qwen2.5:32b	0.917	0.750	0.917	0.917	1.000	−0.083
deepseek-r1:14b	0.889	0.917	0.917	0.912	1.000	−0.083
llama3.1:8b	0.778	0.583	0.750	0.771	0.889	−0.139
qwen2.5:7b	0.861	0.583	0.639	0.736	0.722	−0.083
mistral:7b	0.750	0.417	0.639	0.685	0.722	−0.083
gemma2:27b	0.861	0.583	0.528	0.667	0.611	−0.083
Δ	0.167	0.500	0.389	0.250	0.389	—

Table 13. EMC-lite (N=7, hardened protocol). Source attribution shows strongest differentiation (spread = 0.500).

EMC-lite provides preliminary support for the source-conflict slice. It does not establish EMC as a whole; single-seed evidence only. τ(WMF-AM, EMC-lite) = 0.250: substantial independence.

L MCC-CE-v2 Results

Table 14. MCC-CE-v2 (N=8). Error rates 23–40% (floor eliminated); flagging near zero for all models.

	Balanced (n=30)		Trap-only (n=20)	
Model	Error%	Flag%	Error%	Flag%
qwen2.5:7b	26.7	0.0	40.0	0.0
qwen2.5:14b	36.7	0.0	40.0	0.0
qwen2.5:32b	30.0	0.0	25.0	0.0
llama3.1:8b	26.7	0.0	40.0	0.0
llama3.1:70b	—	—	25.0	0.0
gemma2:27b	30.0	11.1	30.0	0.0
deepseek-r1:14b	23.3	0.0	35.0	0.0
mistral:7b	30.0	0.0	40.0	0.0

The v1 floor was an artifact of easy questions; v2 models make many errors but show near-zero self-flagging under the binary protocol. Alternative designs (continuous confidence, forced ranking) might reveal partial capacity.

M Negative OOD Result

Two OOD stress-test designs (v1: 5-step NL tasks; v2: K=8–9 interference tasks) produced anti-correlated CEF–OOD relationships across model pairs. We make no predictive claim. Full assessment requires $N \geq 11$ with process-sensitive downstream tasks.

N Full Validity Analyses

The validity summary table appears in the main text (Table 6). Below we provide additional details.

Convergent validity details. AGENT-PQ: 10 multi-step scenarios scored by LLM judge (GPT-4o) on 4 rubric dimensions. CONV-RFPOC: 15 reasoning problems with counterfactual chain perturbation. RFPOC = $1 - P$(same answer | perturbed chain). Full protocol in Appendix O. Limitation: LLM judge may introduce shared bias.

Construct separation (WMF-AM $\leftrightarrow$ CLA-DC). $\tau = 0.905$: no evidence of separability at N=7. Both involve state tracking under load. Planned discriminant test at $N \geq 15$. CLA-CR-v3 provides a structurally independent CLA measure. Even if CLA-DC merges with WMF-AM, WMF-AM, MCC-MA, and EMC-lite remain separable ($\tau \leq 0.250$).

Sensitivity. LOMO for CONV-RFPOC vs AGENT-PQ: τ ranges 0.600–1.000 (mean 0.833). Leave-one-family-out (Qwen, N=4): $\tau = 0.667$. After Holm correction, RFPOC–AGENT-PQ remains significant ($p_{\text{adj}} = 0.021$).

Multiple testing. Three primary hypotheses; all others exploratory.

Benchmark governance. All headline results use frozen protocol versions. Probe revisions were motivated by floor/ceiling effects, not model-specific tuning.

O AGENT-PQ Evaluation Protocol

AGENT-PQ uses 10 multi-step scenarios where each model operates within a tool-use scaffold. A separate LLM judge (GPT-4o, temperature = 0) scores each trajectory on four rubric dimensions: plan coherence, state tracking, error recovery, goal adherence. AGENT-PQ = mean of 4 rubric scores/4. AGENT-TC is the binary success rate on the same scenarios.

CONV-RFPOC administers 15 reasoning problems with counterfactual chain perturbation. RFPOC = $1 - P$(same answer | perturbed chain).

P Complete Experimental Protocols

Full prompt templates, parameter specifications, scoring procedures, and reference implementations for all 11 sub-dimensions are provided in the supplementary materials.

Q Weight Sensitivity Analysis

Three weighting schemes (proposed, equal, PCA-derived) are evaluated. Per-dimension profiles are primary; composites are exploratory only.

R Dissociation Case Studies

Matched dissociation pairs: deepseek-r1:14b and qwen2.5:14b share similar EMC-lite (0.912 vs. 0.917) but differ on WMF-AM (0.983 vs. 0.467). mistral:7b and llama3.1:8b have similar WMF-AM but differ on EMC-lite. These cross-dimension pairs anchor the multiple realizability argument.

S Worked Example: WMF-AM at Depth K=3

Setup. "A warehouse has 3 items: Widget=50, Gadget=30, Sprocket=20." Three updates: "Ship 10 Widgets," "Receive 25 Gadgets," "Transfer 5 Sprockets to overflow."
Correct. Widget=40, Gadget=55, Sprocket=15.
Scoring. Exact match per item. deepseek-r1:14b scores 1.000 at K=7; llama3.1:8b scores 0.050. Both score 0.90–0.95 on outcome correctness. This is the Completion Fallacy in action.

T Novelty Classification

Table 15. Novelty tier for all 11 sub-dimensions. Tier 1 = novel; Tier 2 = novel reframing.

Sub-dim	Tier	Prior art / differentiation
WMF-AM	1	Entity Tracking [Kim and Schuster, 2023]: passive retrieval; Latent State Persistence [Anonymous, 2025]: structural ops, no arithmetic load, no agent battery; WMF-AM = cumulative arithmetic transformation + ablation suite + downstream validity
WMF-IM	2	Wang & Sun [Wang and Sun, 2025]: key-value pairs; WMF-IM = NL multi-turn
WMF-IR	2	Extended to domain-varied materials
MCC-MA	1	Gnosis [Ghasemabadi and Niu, 2025] requires internals; MCC-MA = prompting only
MCC-CE	1	Reflexion [Shinn et al., 2023]: aggregate; MCC-CE decomposes detection/correction
EMC-TO	1	No prior LLM temporal ordering measurement
EMC-EI	1	No prior cross-episode interference diagnostic
EMC-SA	2	Interference-conditioned attribution
CLA-DC	1	Intell. Degradation [Wang et al., 2026]: magnitude; CLA-DC = shape
CLA-RA	1	Not previously measured
CLA-CR	1	Not previously measured